Indira Devi Siripurapu

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Education

Ph.D. in Computer Science Spring 2022 – Expected Spring 2027
University of North Texas, Advisor: Dr. Saraju P. Mohanty *Denton, TX*

Dissertation: Advancing Intelligent and Personalized Dietary Systems for Smart Healthcare

Post Graduate Diploma in Business Management (Information Technology and Systems) Jul. 2020 – May 2023
NMIMS Global Access School for Continuing Education, Open and Distance Learning (ODL) *India*

Bachelor of Technology in Electronics and Communication Engineering 2015 – 2019
Jawaharlal Nehru Technological University Hyderabad (JNTUH) *Hyderabad, India*

Completed at Geethanjali College of Engineering and Technology, Keesara

Research Interests

- Artificial Intelligence and Machine Learning; Smart Healthcare; Personalized Nutrition; Food Computing; Computer Vision; Trustworthy and Explainable AI; Privacy-Preserving Machine Learning; Internet of Medical Things; Edge Intelligence; Intelligent Decision-Support Systems; Food–Medication and Herb–Medication Interaction Analysis.

Professional and Research Experience

Doctoral Researcher Spring 2022 – Present
University of North Texas, Smart Electronic Systems Laboratory (SESL) *Denton, TX*

- Developed AI-driven systems for intelligent decision support, predictive modeling, food and image analysis, personalized recommendations, and smart-health applications.
- Designed computer-vision and data-analysis pipelines involving image segmentation, depth estimation, structured and sensor-derived data, and quantitative feature extraction.
- Built privacy-aware and explainable machine-learning frameworks incorporating federated learning, model interpretation, feature engineering, and robustness evaluation.
- Investigated dietary–medication and herb–medication interactions using structured evidence, semantic and chemical representations, and risk-aware prediction methods.
- Designed and evaluated predictive models using controlled test protocols, cross-validation, cold-start evaluation, scalability analysis, and edge-device deployment.
- Contributed to peer-reviewed journal and conference publications in Artificial Intelligence, Machine Learning, Computer Vision, Smart Healthcare, and Edge Intelligence.

Oracle Database Administrator Aug. 2019 – Oct. 2021
Mphasis Pvt. Ltd. *India*

- Progressed to a higher-responsibility role and led major Oracle 11g-to-12c database migration workstreams for enterprise banking environments.
- Served as primary resource contact for 12 projects spanning more than 15 databases and secondary resource on 8 additional projects.
- Led migration and go-live activities for databases exceeding 2 TB, including planning, validation, cutover, and production support.
- Developed and maintained SQL/PLSQL queries and shell scripts and supported database performance, backup and recovery, monitoring, replication, and enterprise database administration.
- Worked with Oracle Enterprise Manager, Data Pump, Data Guard, Transportable Tablespaces, and Oracle GoldenGate.
- Coordinated with onsite teams and business analysts supporting U.S. banking clients JPMorgan Chase and Charles Schwab.

Teaching Experience

Graduate Teaching Assistant Fall 2022 – Present
University of North Texas *Denton, TX*

CSCE 1030 – Computer Science I Fall 2022; Spring 2023; Fall 2023; Fall 2024

- Conducted C/C++ programming laboratories and recitation sessions; developed instructional slides, example programs, and starter code.
- Assisted students with program design, debugging, syntax, algorithmic problem solving, and implementation.
- Graded coursework and examinations, proctored assessments, and held regular office hours.

CSCE 5610 – Computer System Architecture Spring 2024

- Supported course concepts, technical problem solving, grading, assessment administration, and student office hours.

CSCE 3730 – Reconfigurable Logic Fall 2024

- Supported technical coursework and projects, graded assignments and examinations, proctored assessments, and held office hours.

CSCE 3020 – Communications Systems Spring 2025

- Supported quantitative problem solving, graded coursework and examinations, proctored assessments, and held office hours.

CSCE 4901 – Software Development Capstone I Fall 2025; Fall 2026

- Supported senior software-development teams in project planning, requirements analysis, implementation, milestone tracking, and project documentation.
- Evaluated individual and team deliverables and provided project guidance through office hours.

CSCE 4902 – Software Development Capstone II Spring 2026

- Supported teams through software development, testing, refinement, and project completion.
- Evaluated milestones, project documentation, and final deliverables.

Instructional Assistant Summer 2026
University of North Texas *Denton, TX*

CSCE 2100.401 – Foundations of Computing Summer 2026

- Graded coursework using established rubrics and held scheduled office hours for student support.

Publications

Published

- S. S. H. Chakravarthula, I. D. Siripurapu, L. Rachakonda, S. P. Mohanty, and E. Kougianos, "iMediFood-Shield: Secure Edge AI for Food and Medication Interaction Screening," *Electronics*, vol. 15, Art. no. 3799, 2026. DOI: 10.3390/electronics15173799.
- I. D. Siripurapu, L. Rachakonda, S. P. Mohanty, and E. Kougianos, "iLog 2.2: Volume and Nutrition Estimation for Mixed Foods via Mask R-CNN and Federated Learning," *Electronics*, vol. 15, no. 7, Art. no. 1460, 2026. DOI: 10.3390/electronics15071460.
- I. D. Siripurapu, A. Mitra, S. P. Mohanty, and E. Kougianos, "iLog 3.0: Estimating Food Volume from 2D Images Using Mask R-CNN and Monocular Depth Estimation," *2025 IEEE Computer Society Annual Symposium on VLSI*, pp. 1–6, 2025. DOI: 10.1109/ISVLSI65124.2025.11130217.
- I. D. Siripurapu, L. Rachakonda, S. P. Mohanty, and E. Kougianos, "iLog 2.1: Calorie Estimation for Pizza via Mask R-CNN and Federated Learning," *IEEE MetroCon*, 2025. DOI: 10.1109/Metrocon66796.2025.11405824.

Accepted

- I. D. Siripurapu, L. Rachakonda, S. P. Mohanty, and E. Kougianos, "iTexture 2.0: Texture Aware Food and Nutrient Recommendation for Texture Sensitive Toddlers," accepted for IEEE ISVLSI 2026.
- I. D. Siripurapu, L. Rachakonda, S. P. Mohanty, and E. Kougianos, "iGlyHb: IoMT-Driven Glycemic Risk Stratification Using CGM and Dietary Logs," accepted for IEEE ISVLSI 2026.
- I. D. Siripurapu, L. Rachakonda, S. P. Mohanty, and E. Kougianos, "MacroGLU: Explainable and Privacy-Aware Edge Intelligence for Meal-Level Postprandial Glucose Response Modeling," accepted for IFIP 2026.
- I. D. Siripurapu, S. S. H. Chakravarthula, L. Rachakonda, S. P. Mohanty, and E. Kougianos, "iMediFood-AI: Risk-Aware Dietary–Medication Interaction Prediction," accepted for 24th OITS International Conference on Information

Technology (OCIT 2026), subject to satisfactory camera-ready revision.

- I. D. Siripurapu, L. Rachakonda, S. P. Mohanty, and E. Kougianos, "TextureLysis: A Lightweight Framework for Texture-Aware Toddler Feeding," accepted for the 2026 IEEE International Symposium on Smart Electronic Systems (iSES 2026).

Submitted / Under Review

- I. D. Siripurapu, S. S. H. Chakravarthula, L. Rachakonda, and S. P. Mohanty, "iMediFood-MESH: Medication-Supplement Evidential Screening with Hypergraphs," submitted to the 45th IEEE International Conference on Consumer Electronics (ICCE 2027).

Poster Presentations

North Texas Symposium on Generative AI March 7, 2025
University of Texas at Arlington, Arlington, TX

- Poster: "Estimating Food Volume from 2D Images Using Mask R-CNN and Monocular Depth Estimation."

IEEE MetroCon 2025 November 12, 2025

- Poster: "iLog 2.1: Calorie Estimation for Pizza via Mask R-CNN and Federated Learning."
- Awarded **Second Place** in the Graduate Poster Presentation Competition and received a $500 cash prize.

Ongoing Research Directions

- Extend personalized dietary decision support through tighter integration of food composition, glucose response, behavioral context, and medication-aware nutrition.
- Develop larger and more externally validated glucose-aware nutrition models with stronger personalization, explainability, and edge/mobile deployment.
- Advance texture-aware feeding support using longitudinal caregiver data, adaptive preference learning, and human-centered recommendation strategies.
- Expand trustworthy dietary–medication screening through improved evidence integration, uncertainty handling, and clinically oriented validation.

Journal / Conference Reviewer

- *Integration, the VLSI Journal*, 2026. Reviewed manuscript: "A Side-Channel Attack on Reinforcement Learning Models" (Manuscript VLSIJ-D-26-00551R2).
- IEEE Computer Society Annual Symposium on VLSI, 2026.
- IEEE International Symposium on Smart Electronic Systems, 2026.
- IFIP Internet of Things Conference, 2026.
- OITS International Conference on Information Technology, 2026.

Technical Skills

Programming: Python, Java, C, C++, JavaScript, SQL, PL/SQL, MATLAB

Deep Learning & AI: Neural Networks, CNNs, LSTMs, Transformers, Large Language Models (LLMs), Computer Vision, Natural Language Processing, Explainable AI, Federated Learning

Tools & ML Frameworks: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, PySpark, NLTK, Matplotlib, Jupyter, Google Colab, PyCharm, Visual Studio Code, Git

Software Development: Django, Spring Boot, Node.js, REST APIs, Agile/Scrum, Debugging, Testing, Technical Documentation

Databases & Cloud: Oracle, MySQL, SQL Server, MongoDB, DynamoDB, AWS, Microsoft Azure, Google Cloud Platform (GCP)

Enterprise & Systems: Oracle Database Administration, Linux, Edge Computing, IoT/IoMT, Raspberry Pi

Internships

Microsoft Technology Associate (MTA) Program May – June 2018
Verzeo Edutech *Hyderabad, India*

- Applied Python and introductory machine-learning workflows involving classification, regression, and data analysis.

Research Intern June – August 2018
Electronics Corporation of India Limited (ECIL) *Hyderabad, India*

- Worked on the Radiation & Gas Monitoring System (RGMS) and gained hands-on exposure to instrumentation and

monitoring systems used in defense-oriented environments, including naval applications.

Undergraduate Intern May – June 2017
ECIL Training Center *Hyderabad, India*

- Completed VLSI/UART training and developed a wireless RF-controlled robot using a microcontroller.

Awards and Honors

- **Second Place, Graduate Poster Presentation Competition**, IEEE MetroCon 2025 – $500 cash prize.
- Tuition Benefit Scholarship, Toulouse Graduate School, University of North Texas, Fall 2022 – Present.
- Golden Key International Honour Society.
- Employee of the Month, Mphasis Pvt. Ltd., September 2020 and November 2020.
- Pandemic Warrior Award, Mphasis Pvt. Ltd., 2020.
- First Place, Brain Tumor Extraction Project, National TechExpo, 2019.
- Business English Certificate Preliminary, Cambridge English.

Professional Development and Memberships

- Participant, University of North Texas SEED Program, Fall 2026.
- IEEE Student Member.
- IEEE Young Professionals.

Leadership and Service

Head of Cultural Department Spring 2022; Fall 2022
Indian Student Association, University of North Texas *Denton, TX*

- Coordinated student arrival support, airport rides, enrollment and campus guidance, cultural events, festival celebrations, and donation initiatives.

President, National Service Scheme (NSS) 2015 – 2019
Geethanjali College of Engineering and Technology *Hyderabad, India*

- Coordinated village outreach, blood-donation and dental camps, digital-literacy initiatives, food and blanket drives, emergency relief, education-awareness, and women's-rights initiatives.

Student Secretary, IEEE Student Chapter 2015 – 2019
Geethanjali College of Engineering and Technology *Hyderabad, India*

- Coordinated technical events, workshops, seminars, student participation, communications, and chapter logistics.

Campus Placement Volunteer / Coordinator 2016 – 2019
Geethanjali College of Engineering and Technology *Hyderabad, India*

- Coordinated campus recruitment logistics, students, visiting companies, and communication with HR representatives and company leadership.